

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I Vasantha Nithi D Y (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough → Possible Flu

- li. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- lii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- lv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

- (a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.
- (b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.
- (c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 – Smart Medical Diagnosis System

Problem Understanding

The hospital uses a rule-based logical agent for preliminary diagnosis. The knowledge base contains rules connecting symptoms to possible diseases. For **Patient A**, the known facts are:

- Fever = True
- Cough = True
- Breathlessness = True

The system must use **Propositional Logic, Forward Chaining, and Backward Chaining** to determine the possible diagnoses and verify whether Patient A has Pneumonia.

(a) Propositional Logic Representation

Define the Symbols

Symbol Meaning

F	Patient has Fever
C	Patient has Cough
R	Patient has Rash
B	Patient has Breathlessness
Flu	Patient has Possible Flu
M	Patient has Possible Measles
P	Patient has Possible Pneumonia

Rules

Rule 1:

If the patient has Fever **AND** Cough, then Possible Flu.

Rule 2:

If the patient has Fever **AND** Rash, then Possible Measles.

Rule 3:

If the patient has Cough **AND** Breathlessness, then Possible Pneumonia.

Facts for Patient A

Therefore, the knowledge base contains:

(b) Forward Chaining**Method**

Forward Chaining starts with the known facts and repeatedly applies rules whose conditions are satisfied. New conclusions are added to the knowledge base.

Inference Step 1

Known facts:

Check **Rule 1**:

Since both **F** and **C** are true:

Rule 1 is fired.

Inference Step 2

Check **Rule 2**:

We know:

but there is no fact that:

Therefore, **Rule 2 cannot be fired.**

No Measles diagnosis is derived.

Inference Step 3

Check **Rule 3**:

Since:

and

both conditions are satisfied.

Therefore:

Rule 3 is fired.

Forward Chaining Table

Step	Rule	Checked Conditions	Derived Fact
1	Rule 1	$F \wedge C = \text{True}$	Flu
2	Rule 2	$F \wedge R = \text{False}$	None
3	Rule 3	$C \wedge B = \text{True}$	Pneumonia

All Possible Diagnoses

Therefore, the system derives:

and

Measles cannot be derived because the patient has no known Rash.

(c) Backward Chaining for Pneumonia

Goal

The goal is:

Backward Chaining starts with the goal and works backward to determine whether its required conditions can be established.

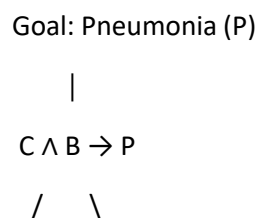
Goal Reduction

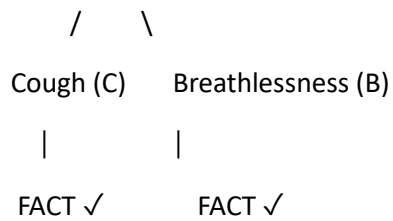
The rule that produces Pneumonia is:

Therefore, to prove **P**, we need:

and

Goal Reduction Tree





Both required sub-goals are already present in Patient A's facts.

Therefore:

Hence:

Analysis

The two reasoning methods reach the same conclusion but work in opposite directions:

- **Forward Chaining:** starts from symptoms and derives possible diagnoses.
- **Backward Chaining:** starts from the desired diagnosis and checks whether its required conditions are satisfied.
- Patient A satisfies the conditions for both **Flu** and **Pneumonia**.
- The condition for **Measles** is not satisfied because Rash is not present.

Thus, the logical agent can confirm **Possible Pneumonia** from the available knowledge base.

Result

Using Propositional Logic, Forward Chaining, and Backward Chaining, the system successfully derives:

The goal **"Patient A has Pneumonia"** is successfully verified because both **Cough** and **Breathlessness** are known facts.

Conclusion

The Smart Medical Diagnosis System successfully demonstrates knowledge representation and logical reasoning. Forward Chaining derives diagnoses from known symptoms, while Backward Chaining verifies a specific diagnosis by reducing it to required conditions. For Patient A, the available facts logically support **Possible Flu and Possible Pneumonia**, while **Measles cannot be concluded** from the given knowledge base.

Case Study 2 – Autonomous Traffic Management Agent

Problem Understanding

The smart-city traffic controller uses a First-Order Logic (FOL) knowledge base to manage emergency vehicle routing. The system uses Unification and Resolution to determine whether CarA can proceed behind an ambulance.

Knowledge Base

Rules

Rule 1:

For every vehicle x , if $\text{Vehicle}(x)$ AND $\text{EmergencyType}(x)$, then $\text{ClearPath}(x)$.

Rule 2:

For every vehicle x , if $\text{ClearPath}(x)$, then $\text{GreenSignal}(x)$ AND NOT $\text{RedSignal}(x)$.

Rule 3:

For every x and y , if $\text{Vehicle}(x)$ AND $\text{Behind}(x,y)$ AND $\text{GreenSignal}(y)$, then $\text{Proceed}(x)$.

Facts

- $\text{Vehicle}(\text{Ambulance})$
 - $\text{EmergencyType}(\text{Ambulance})$
 - $\text{Vehicle}(\text{CarA})$
 - $\text{Behind}(\text{CarA}, \text{Ambulance})$
-

(a) Apply Unification

Rule 1

Rule 1 contains the variable x :

$\text{Vehicle}(x)$ AND $\text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

The given facts are:

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

Therefore, x is replaced by Ambulance.

Substitution:

$x = \text{Ambulance}$

Hence:

$\text{ClearPath}(\text{Ambulance})$

is derived.

Rule 2

Rule 2 is:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x)$

Since we have:

$\text{ClearPath}(\text{Ambulance})$

we substitute:

$x = \text{Ambulance}$

Therefore:

$\text{GreenSignal}(\text{Ambulance})$

is derived.

Unification Summary

Step	Rule	Substitution	Derived Fact
1	Rule 1	$x = \text{Ambulance}$	$\text{ClearPath}(\text{Ambulance})$
2	Rule 2	$x = \text{Ambulance}$	$\text{GreenSignal}(\text{Ambulance})$

(b) Resolution Method

To prove:

Proceed(CarA)

Convert Rule 1 to CNF

Original:

$\text{Vehicle}(x) \text{ AND } \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

CNF:

NOT Vehicle(x) OR NOT EmergencyType(x) OR ClearPath(x)

Convert Rule 2 to CNF

CNF:

NOT ClearPath(x) OR GreenSignal(x)

Convert Rule 3 to CNF

Original:

Vehicle(x) AND Behind(x,y) AND GreenSignal(y) $\rightarrow$ Proceed(x)

CNF:

NOT Vehicle(x) OR NOT Behind(x,y) OR NOT GreenSignal(y) OR Proceed(x)

Resolution Steps

From Rule 1 and the facts:

Vehicle(Ambulance)

EmergencyType(Ambulance)

we derive:

ClearPath(Ambulance)

From Rule 2:

ClearPath(Ambulance)

we derive:

GreenSignal(Ambulance)

Now consider Rule 3.

Given:

Vehicle(CarA)

Behind(CarA, Ambulance)

GreenSignal(Ambulance)

All the required conditions are satisfied.

Therefore:

Proceed(CarA)

is derived.

Resolution Chain

Vehicle(Ambulance) + EmergencyType(Ambulance)

↓

ClearPath(Ambulance)

↓

GreenSignal(Ambulance)

↓

Vehicle(CarA) + Behind(CarA, Ambulance) + GreenSignal(Ambulance)

↓

Proceed(CarA)

(c) Two Simultaneous Emergency Vehicles

The current knowledge base can handle multiple emergency vehicles because Rule 1 applies to every vehicle.

For example, if another emergency vehicle is introduced, the following facts can be added:

Vehicle(Ambulance2)

EmergencyType(Ambulance2)

The system can then derive:

ClearPath(Ambulance2)

and:

GreenSignal(Ambulance2)

However, if two emergency vehicles require the same route, additional rules are needed to handle priority and route conflicts.

Required Additional Information

- Emergency status of each vehicle
- Position of each vehicle

- Route or path requirements
- Priority rule for multiple emergency vehicles

This would make the traffic management system safer and more effective.

Analysis

The FOL knowledge base successfully connects emergency vehicle status with traffic signals and vehicle movement. Unification identifies the correct substitutions, while Resolution allows the system to derive new facts logically.

Result

The required conclusions are:

1. ClearPath(Ambulance)
2. GreenSignal(Ambulance)
3. Proceed(CarA)

Therefore, the system successfully proves that **CarA can proceed behind the ambulance**.

Conclusion

Thus, the Autonomous Traffic Management Agent successfully applies FOL, Unification, and Resolution to reason about emergency vehicle routing. The system determines that CarA can proceed based on the available facts and rules.

Case Study 3 – Agricultural Expert Reasoning System

Problem Understanding

The agricultural advisory system uses a **propositional knowledge base** to determine whether irrigation is required and whether the drip method should be applied. The given facts are **SoilDry = True** and **CropWheat = True**. The task is to convert the knowledge base into CNF, prove **ApplyDripMethod** using Resolution Refutation, and analyze what happens if SoilDry is removed.

(a) Conversion to CNF

P1

Original:

SoilDry $\rightarrow$ IrrigationNeeded

Eliminate implication:

NOT SoilDry OR IrrigationNeeded

So the CNF is:

NOT SoilDry OR IrrigationNeeded

P2

Original:

IrrigationNeeded AND CropWheat $\rightarrow$ ApplyDripMethod

Eliminate implication:

NOT (IrrigationNeeded AND CropWheat) OR ApplyDripMethod

Using De Morgan's law:

NOT IrrigationNeeded OR NOT CropWheat OR ApplyDripMethod

So the CNF is:

NOT IrrigationNeeded OR NOT CropWheat OR ApplyDripMethod

P3

Original:

NOT ApplyDripMethod AND CropWheat $\rightarrow$ CropAtRisk

Eliminate implication:

NOT (NOT ApplyDripMethod AND CropWheat) OR CropAtRisk

Using De Morgan's law:

ApplyDripMethod OR NOT CropWheat OR CropAtRisk

So the CNF is:

ApplyDripMethod OR NOT CropWheat OR CropAtRisk

Facts

SoilDry = True

CropWheat = True

Therefore:

SoilDry

CropWheat

are added directly to the CNF knowledge base.

(b) Resolution Refutation

Goal

We need to prove:

ApplyDripMethod

For Resolution Refutation, negate the goal:

NOT ApplyDripMethod

Add this to the knowledge base.

Resolution Step 1

From P1:

NOT SoilDry OR IrrigationNeeded

and the fact:

SoilDry

we derive:

IrrigationNeeded

Resolution Step 2

From P2:

NOT IrrigationNeeded OR NOT CropWheat OR ApplyDripMethod

Using:

IrrigationNeeded

and:

CropWheat

we derive:

ApplyDripMethod

Resolution Step 3

We already have the negated goal:

NOT ApplyDripMethod

Combine it with:

ApplyDripMethod

This produces:

Empty Clause ($\square$)

Therefore, the negation of the goal leads to a contradiction.

Hence:

ApplyDripMethod is proved.

Resolution Proof

SoilDry

+

NOT SoilDry OR IrrigationNeeded

↓

IrrigationNeeded

IrrigationNeeded + CropWheat

+

NOT IrrigationNeeded OR NOT CropWheat OR ApplyDripMethod



ApplyDripMethod

ApplyDripMethod + NOT ApplyDripMethod



Empty Clause (□)

(c) Removing SoilDry

If the fact:

SoilDry

is removed, Rule P1 cannot be triggered.

Therefore:

IrrigationNeeded

cannot be derived.

Since P2 requires both:

IrrigationNeeded AND CropWheat

the system can no longer derive:

ApplyDripMethod

However, **CropAtRisk** also does not follow from the remaining facts alone.

P3 requires:

NOT ApplyDripMethod AND CropWheat

Although CropWheat is still true, there is no explicit fact proving **NOT ApplyDripMethod**.

Therefore:

CropAtRisk cannot be logically concluded.

Analysis

Removing SoilDry breaks the reasoning chain:

SoilDry → IrrigationNeeded → ApplyDripMethod

Without SoilDry, irrigation cannot be established, so ApplyDripMethod cannot be proven. At the same time, the absence of ApplyDripMethod is not automatically equivalent to NOT ApplyDripMethod.

Result

With SoilDry present:

SoilDry → IrrigationNeeded → ApplyDripMethod

Therefore, **ApplyDripMethod** is proved.

Without SoilDry, **ApplyDripMethod cannot be derived**, and **CropAtRisk also does not logically follow** from the remaining facts.

Conclusion

Thus, CNF conversion and Resolution Refutation successfully prove **ApplyDripMethod** when SoilDry and CropWheat are known facts. Removing SoilDry prevents the irrigation and drip-method conclusions from being derived.

Case Study 4 – Wumpus World Knowledge Agent

Problem Understanding

The Wumpus World agent receives three percepts:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

Using the given propositional rules, the agent must construct its belief state, identify possible Wumpus and Pit locations, and determine whether the path **[1,1] → [1,2] → [2,2]** is safe.

(a) Belief State and Modus Ponens

Given Percepts

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Rule 1

Stench[1,2] $\rightarrow$ Wumpus is adjacent to [1,2]

Since Stench[1,2] is true:

Wumpus is adjacent to [1,2]

is derived.

Rule 2

Breeze[1,1] $\rightarrow$ Pit is adjacent to [1,1]

Since Breeze[1,1] is true:

Pit is adjacent to [1,1]

is derived.

Rule 3

Glitter[x,y] $\rightarrow$ Gold is at [x,y]

Since Glitter[2,2] is true:

Gold is at [2,2]

is derived.

Derived Knowledge

Percept	Rule Applied	Conclusion
Stench[1,2]	Rule 1	Wumpus adjacent to [1,2]
Breeze[1,1]	Rule 2	Pit adjacent to [1,1]
Glitter[2,2]	Rule 3	Gold at [2,2]

(b) Possible Wumpus and Pit Locations

Wumpus

The stench at **[1,2]** indicates that the Wumpus is adjacent to [1,2].

Therefore, possible Wumpus locations are the neighboring cells of [1,2], such as:

[1,1], [1,3], [2,2]

subject to the available grid boundaries.

Pit

The breeze at [1,1] indicates that a Pit is adjacent to [1,1].

Therefore, possible Pit locations are:

[1,2], [2,1]

Forward Chaining

Stench[1,2]

↓

Wumpus adjacent to [1,2]

Breeze[1,1]

↓

Pit adjacent to [1,1]

Glitter[2,2]

↓

Gold at [2,2]

Thus, the agent cannot identify the exact Wumpus or Pit location from the given percepts alone.

(c) Safety of the Proposed Path

Proposed Path

[1,1] → [1,2] → [2,2]

The path cannot be guaranteed to be safe.

The reason is that:

- **[1,1]** has a Breeze, indicating a possible Pit nearby.

- **[1,2]** has a Stench, indicating a possible Wumpus nearby.
- The given knowledge does not establish that **[1,2]** or **[2,2]** is definitely free from hazards.
- Glitter at **[2,2]** confirms that gold is present there, but it does **not** prove that the cell is safe.

Therefore, the agent should **not assume the complete path is safe** based only on the available knowledge.

Analysis

The agent can derive the presence of possible hazards from its percepts, but the given rules do not provide enough information to determine their exact locations. Therefore, logical reasoning identifies **possible danger**, rather than guaranteeing a safe path.

Result

The agent successfully derives:

- **Wumpus:** Possible locations adjacent to **[1,2]**
- **Pit:** Possible locations adjacent to **[1,1]**
- **Gold:** Located at **[2,2]**

The proposed path **[1,1] → [1,2] → [2,2]** **cannot be confirmed as completely safe**.

Conclusion

Thus, the Wumpus World knowledge agent successfully uses **Modus Ponens and Forward Chaining** to update its belief state. However, because the available knowledge does not eliminate the possible Wumpus and Pit locations, the agent should proceed cautiously rather than assume the proposed path is safe.